

ARTHUR PEREIRA VIEGAS

AI Engineer · RAG, Multi-Agent Systems & LLM Infrastructure

Porto Alegre, Brazil · Remote (UTC-3), overlapping US & EU business hours · Open to contractor roles
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SUMMARY

Software Engineer with 4 years of experience building production AI systems: end-to-end RAG pipelines, multi-agent orchestration with LangGraph, MCP servers, and cloud-native backends on AWS. At Dataglass, led the full Dockerization and CI/CD automation of a production healthcare-data platform — eliminating 100% of manual deploy steps and cutting environment setup from 2+ hours to under 5 minutes. Ships systems with the architecture decisions written down: eight ADRs published alongside a modular Go monolith.

PROFESSIONAL EXPERIENCE

- Full-Stack Developer, Dataglass

Jan 2024 – Present

 - **Technical Leadership:** Led development of the core data-visualization application (Django, React) serving healthcare clients through Tableau Cloud APIs. Mentored a junior engineer via code review and sprint planning, cutting PR review cycles by ~40%.
 - **AI in Product:** Built an in-product AI agent on Vertex AI with Gemini function calling, performing retrieval-augmented Q&A over client data and orchestrating multi-step workflows.
 - **DevOps & Cloud:** Architected and implemented the full Dockerization of the platform, containerizing every service and managing images on AWS ECR — cutting environment setup from 2+ hours to under 5 minutes. Migrated application data and assets to S3, decoupling file storage from the server.
 - **CI/CD Automation:** Built a fully automated pipeline (AWS CodePipeline, CodeBuild, Elastic Beanstalk), eliminating 100% of manual deploy steps and the incidents caused by them. Async processing with Celery + Redis/ElastiCache.
- Full-Stack Development Intern, Dataglass

May 2023 – Dec 2023

 - Progressed from front-end (React) tasks to full-stack ownership of the core Django application, earning an early promotion within 7 months. Built and maintained data-analysis features used daily by healthcare clients.
- Software Developer · Architecture · Project Management — AGES, PUCRS

2021 – 2025

 - Delivered software for real external clients contracting PUCRS's software engineering agency (AGES) across four consecutive year-long cycles — one delivery team per year, with versioned releases and client-facing demos.
 - Held a different role each cycle — hands-on development, technical architecture, and project management — in cross-functional teams running sprint cadence with mandatory code review.
- Community Analysis Intern, DELL Technologies / PUCRS

Apr 2021 – May 2023

Globally distributed team (USA, India) on Dell Brazil's official community forum: technical guides and weekly data-driven reporting to international management.

SELECTED PROJECTS

- Mirante — Go · SvelteKit · libSQL/Turso · SSE · OpenTelemetry · Docker

 - Modular monolith in Go where domain packages never import each other — cross-domain access goes through ports and typed IDs. Real-time push over SSE, migrations embedded via go:embed, per-user data isolation, OTLP observability. Eight ADRs published: github.com/ArthurViegas01/mirante/tree/main/docs/adr
- Encaixe — FastAPI · LangGraph · Claude · Voyage AI · pgvector · Celery · Next.js · Terraform

 - Multi-tenant WhatsApp customer-service SaaS. A 6-node LangGraph state machine classifies intent, retrieves context via semantic search, books appointments through Google Calendar OAuth2 and escalates to a human on low confidence. Tenant isolation via PostgreSQL row-level security.
- Context — FastAPI · Celery · pgvector · fastembed/ONNX · LangChain · Docker

 - End-to-end RAG pipeline over private documents: async ingestion, chunking, local ONNX embeddings into pgvector (HNSW), and hybrid retrieval fusing vector similarity with Portuguese full-text search through Reciprocal Rank Fusion.
- Devscope — Python · FastMCP · Groq/Llama 3.1-70b · LangChain · Redis · Terraform

 - Model Context Protocol server exposing GitHub profile-analysis tools to LLM clients over Streamable HTTP / JSON-RPC — four FastMCP tools with structured JSON output, Redis caching and a Lua atomic rate limiter.

TECHNICAL SKILLS

- **AI / LLM Engineering:** RAG, LangChain, LangGraph, MCP (Model Context Protocol), hybrid search & Reciprocal Rank Fusion, pgvector (HNSW), embeddings (Voyage AI, fastembed/ONNX), Vertex AI, Gemini, Claude, Groq / Llama 3, Ollama, structured output via JSON Schema
- **Languages:** Python, TypeScript, JavaScript, Go, SQL
- **Back-End:** FastAPI, Django, Django REST Framework, Celery, Node.js, REST APIs, modular monolith & hexagonal architecture, SSE
- **Front-End:** React, Next.js, Vue 3, SvelteKit, Tailwind CSS, SASS
- **Cloud & DevOps:** AWS (CodePipeline, CodeBuild, Elastic Beanstalk, EC2, S3, ECR, RDS), GCP, Docker, Terraform, CI/CD, GitHub Actions, OpenTelemetry
- **Databases:** PostgreSQL, pgvector, Redis, ElastiCache, SQLite / libSQL, Supabase
- **Practices:** Architecture Decision Records, code review, testing (pytest, Go testing, testcontainers), Git Flow, Agile / Scrum
- **Spoken Languages:** Portuguese (native) · English (advanced / professional)

EDUCATION

B.Sc. in Software Engineering — PUCRS, Porto Alegre, Brazil Jan 2021 – Jun 2025